Rave Generation

KNOCK

Ultimate Techno Bassdrum

User Manual

Overview

Knock is a TR-909-based bassdrum synthesizer with transient shaping, analog-modeled drive, and a built-in reverb send. Designed for electronic music production, it delivers everything from classic house kicks to hard-hitting techno bassdrums in a single plugin.

Features include dual oscillator modes (Classic and ALT), 909-style click generator, adjustable body and snap transients, a dedicated Punch FX section with multiple filter types and distortion algorithms, one-knob output drive with coloration, Clip 0dB protection, and a built-in RRV-10 reverb send with 9 modes.

Signal Flow

MIDI Note → 909 Oscillator → Click/Body/Snap Transients → Punch FX → Output Drive + Coloration → Reverb Send → Smart Clip

The bassdrum signal starts with the oscillator (Classic or ALT mode), shaped by attack, decay, and hold envelopes. Transient layers (click, body, snap) are added on top. The signal then passes through the Punch FX section for additional character, followed by output drive and coloration filters. Finally, an additive reverb send adds space, and Clip 0dB ensures clean output.

Default
KNOCK
CORE
TUNE
ATTACK
DECAY
HOLD
BODY
SNAP
LEVEL
0.500
0.500
0.500
0.580
0.000
0.000
0.500
OSCILLATOR
< Classic >
CLICK
< Classic >
MIDI
TUNE x2
DECAY x2
PUNCH
TICK
TIME
FREQ
RESO
GRIT
TONE
MIX
0.000
0.300
0.300
0.100
0.000
0.500
1.000
FM TICK
FILTER
< Off >
ALGO
< Tube >
OUTPUT
DRIVE
HI MID
HIGH
OUTPUT
CLIP 0dB
0.000
0.500
0.500
0.500
SEND FX
PRE-DLY
TIME
TONE
DRIVE
SEND
TAIL
0.000
5
0.500
0.000
0.000
Off
REVERB
< Room 1 >

CORE Section

The Core section contains the fundamental 909 bassdrum synthesis engine and transient controls.

Oscillator Controls

TUNE: Controls the pitch sweep of the kick. Higher values produce a longer, more pronounced pitch drop at the start of each hit.

ATTACK: Controls the initial transient loudness. Higher values produce a sharper, more prominent click at the start of the kick.

DECAY: Controls how long the bassdrum rings out. Higher values produce longer, more sustained kicks.

HOLD: Sets the hold time (0–100 ms) before the decay begins. The sound stays at full volume for this duration before fading out. Creates a fatter, more sustained kick at higher values.

BODY: Adds weight and low-end presence to the kick. Crank it up for a heavier, more powerful sound.

SNAP: Controls the snap transient level. Adds a sharp, clicky attack for cut-through in a mix.

LEVEL: Master velocity/level control for the bassdrum output.

Switches

OSCILLATOR: Selects between Classic (authentic 909 triangle tone) and ALT (cleaner, sine-like tone).

CLICK: Selects the click character. Classic gives the original 909 click sound, ALT provides a punchier, harder-edged alternative.

MIDI: Toggles chromatic MIDI note tracking. When ON, the oscillator follows incoming MIDI note pitch, allowing melodic/chromatic playing.

TUNE x2: Doubles the tuning range for higher-pitched kicks.

DECAY x2: Doubles the decay time range for longer, boomy kicks.

PUNCH Section

The Punch section adds 909-style transient processing with filter and distortion effects.

Controls

TICK: Controls the tick transient level, adding a sharp percussive click at the very start of the sound.

TIME: Sets the duration of the punch envelope. Short values create tight snaps, longer values create more drawn-out transient tails.

FREQ: Sets the frequency for the punch filter.

RESO: Filter resonance amount. Higher values create more aggressive, resonant character.

GRIT: Controls the distortion intensity within the punch section.

TONE: Shapes the tonal balance of the punch effect.

MIX: Blends the punch effect with the dry kick signal.

Selectors

FILTER: Selects the punch filter type: Off (bypass), LP (lowpass), HP (highpass), or BP (bandpass).

ALGO: Selects the punch distortion algorithm: Tube (warm valve saturation), Tape (magnetic tape compression), Clip (hard clipping), or Fold (wavefolding).

FM TICK: When enabled, replaces the standard tick transient with an FM-synthesized metallic tick that tracks the oscillator pitch. Produces a sharper, more aggressive transient character.

OUTPUT Section

The Output section provides final signal shaping with drive, coloration, and level control.

Controls

DRIVE: One-knob distortion with a warm, tube-like character. Smoothly goes from clean to crunchy across the full range.

HI MID: Adds or removes midrange presence around 2.4 kHz. Most effective with the Classic oscillator which has richer harmonics.

HIGH: Brightens or darkens the overall tone by adjusting high-frequency content.

OUTPUT: Master output level control.

CLIP 0dB: Enables Clip 0dB protection. Prevents the output from exceeding 0 dB to avoid digital clipping.

SEND FX Section

The Send FX section provides a built-in reverb effect based on the RRV-10 vintage reverb engine.

Controls

PRE-DLY: Pre-delay time before the reverb effect begins. Adds separation between the dry kick and the reverb tail.

TIME: Reverb decay time (stepped 0–15). Controls how long the reverb rings out.

TONE: Shapes the reverb tone. Turn left for darker, warmer tails. Turn right for brighter, more present reverb. Center position is neutral.

DRIVE: Adds drive and saturation to the reverb signal for a grittier, more colored reverb tail.

SEND: Controls the reverb send level. This is additive (not a dry/wet mix), so the dry signal always passes through at full level.

Selectors

REVERB: Selects the reverb algorithm: Room 1, Room 2, Hall 1, Hall 2, Plate 1, Plate 2, M Tap 1, M Tap 2, or Gated. Each mode offers a different spatial character from tight rooms to expansive halls and rhythmic multi-tap delays.

TAIL: Controls how the reverb tail behaves on each new MIDI note trigger. **Off** lets the reverb ring naturally with full buildup between hits. **Reset** clears the reverb buffer on each trigger for a clean tail per hit. **Duck** uses a sidechain envelope follower on the dry kick to duck the reverb tail, letting it swell back up after each hit for a pumping, breathing effect.

Quick Start

Classic 909 kick: Set OSCILLATOR to Classic, TUNE around 0.5, DECAY to taste. Add BODY for weight, SNAP for attack.

Hard techno kick: Drive the OUTPUT DRIVE to 40–60%, enable CLIP 0dB. Use the PUNCH section with LP filter and Tube algo for aggressive character. Set SEND FX TAIL to Reset for clean reverb per hit.

Deep room kick: Set SEND FX to Room 1 or Room 2, TIME around 5–8, SEND at 20–30%. Turn TONE left for a dark, subby reverb tail.

Gated reverb kick: Select Gated reverb mode, TIME at 3–5, SEND at 40–60%. Creates a tight, punchy reverb that cuts off sharply.

Clean sub kick: Use ALT oscillator for a pure sine-like tone. Keep DRIVE and PUNCH minimal. Lower DECAY for a tight sub hit.

Layered transient kick: Combine SNAP, TICK, and BODY for complex transient layering. Use PUNCH TIME to shape the transient envelope length.

Technical Notes

The Classic oscillator is modeled after the original TR-909 bassdrum circuit for authentic sound and dynamics. The ALT oscillator uses a different synthesis method for a cleaner, purer tone.

Velocity response is modeled after the original hardware for authentic dynamics.

The reverb engine is modeled after the BOSS RV-10 for an authentic vintage reverb character.

The output drive is designed for a warm, tube-like character without unwanted volume jumps.

Clip 0dB uses a two-stage approach for transparent limiting that prevents clipping without audible artifacts.

HI MID and HIGH controls are most effective with the Classic oscillator. The ALT oscillator produces a nearly pure tone with fewer harmonics to shape.

Support

For technical support, updates, and additional information: ravegeneration.io